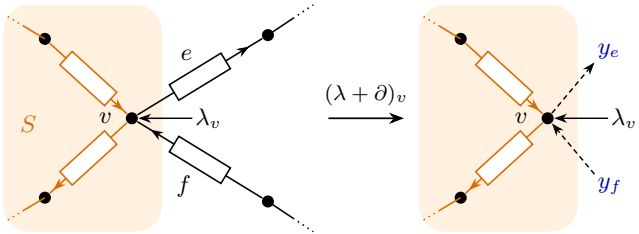
(a) The island is fed at v

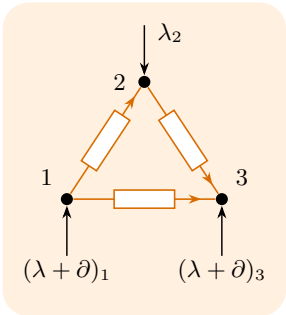


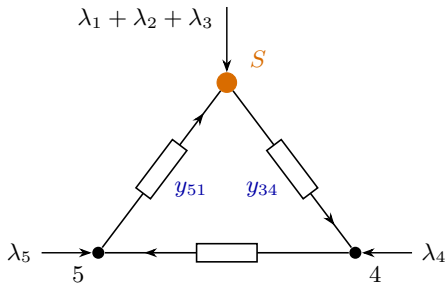
$$(\lambda+\partial)_{v,+\dot{a}} = \lambda_{v,+\dot{a}} + \frac{\partial}{\partial z_e^{+\dot{a}}} - \frac{\partial}{\partial z_f^{+\dot{a}}}$$

$$\frac{\partial}{\partial z_e^{+\dot{a}}}\, e^{-z_e^{+\dot{b}} y_{e,+\dot{b}}} = -y_{e,+\dot{a}}\, e^{-z_e^{+\dot{b}} y_{e,+\dot{b}}}$$

$$(\lambda+\partial)_{v,+\dot{a}} \longmapsto \lambda_{v,+\dot{a}} - y_{e,+\dot{a}} + y_{f,+\dot{a}} = \lambda^{\gamma}_{v,+\dot{a}}$$

(b) The house, $\Gamma_1 = (12, 13, 23, 34, 45, 51)$, at the face of $S = \{1, 2, 3\}$





$\mathcal{I}^{\text{raw cut}}_{\Gamma[S]}(\lambda+\partial; z)$ $\qquad$ $\mathcal{I}^{\text{raw cut}}_{\Gamma(S)}(\lambda; z)$

$$(\lambda+\partial)_{1,+\dot{a}} = \lambda_{1,+\dot{a}} - \frac{\partial}{\partial z_{51}^{+\dot{a}}} \;\longmapsto\; \lambda_{1,+\dot{a}} + y_{51,+\dot{a}}$$

$$(\lambda+\partial)_{2,+\dot{a}} = \lambda_{2,+\dot{a}}$$

$$(\lambda+\partial)_{3,+\dot{a}} = \lambda_{3,+\dot{a}} + \frac{\partial}{\partial z_{34}^{+\dot{a}}} \;\longmapsto\; \lambda_{3,+\dot{a}} - y_{34,+\dot{a}}$$

$$\sum_{v\in S}(\lambda+\partial)_{v,+\dot{a}} \longmapsto (\lambda_1+\lambda_2+\lambda_3)_{+\dot{a}} + y_{51,+\dot{a}} - y_{34,+\dot{a}} = 0$$